

An Unbounded Diagonal Semigroup on the Bourgain–Delbaen Space and the Dual of Its Sun Dual

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Status. Candidate full solution, likely valid, subject to expert review. The construction has an unbounded generator and therefore meets the strongest natural interpretation of “non-trivial” in the source question.

Source and Extracted Question

The source is Philip Preussler and Felix L. Schwenninger, *Implications of structured continuous maximal regularity*, arXiv:2605.12121v1 [1]. In Remark 4.15(2), page 24, the authors consider the classical Bourgain–Delbaen space Y , which satisfies

$$Y \not\supset c_0, \quad Y' \simeq \ell^1, \quad Y'' \simeq \ell^\infty \supset c_0,$$

and ask whether there exists a non-trivial C_0 -semigroup on Y such that

$$(Y^\odot)' \supset c_0.$$

Here $Y^\odot$ is the sun dual associated with the semigroup. The source immediately discusses the possibility that Y has the Lotz property, i.e. that every C_0 -semigroup on Y has bounded generator. Thus the strongest and mathematically substantive reading of “non-trivial” is “having unbounded generator.” The construction below has that property.

2. A result of Bessaga and Pełczyński [15, Thm. 4], see also [70, Prop. 2.e.8] implies that $X^\odot$ contains ℓ^1 as a complemented subspace if and only if $X^{\odot'}$ contains c_0 . The Bourgain–Delbaen space Y , see [17, Ch. 5], cf. also [46, Thm. 487], has the property that $Y \not\supset c_0$ but $Y' \simeq \ell^1$ such that $Y'' \simeq \ell^\infty \supset c_0$. However, we do not know if there is a (non-trivial) C_0 -semigroup on Y such that $Y^{\odot'} \supset c_0$. For example, Y could have the Lotz property, i.e., that every C_0 -semigroup on Y has a bounded generator; cf., [93]. Lotz showed in [71, Thm. 3] that semigroups on Grothendieck spaces with the Dunford–Pettis property (see, e.g., [71, p. 208]) have bounded generators; whence such spaces are Lotz. Regarding the Bourgain–Delbaen space Y : it is known that Y is not a Grothendieck space [43, Ex. 4.3.1] although Y has the Dunford–Pettis property [17, p. 155]. It is also known in the folklore that Y is not a Banach lattice (see, e.g., [89, p. 46]), which means that the characterization of the Lotz property on Banach lattices [93, Thm. 2] cannot be used.

Definitions

Let X be a Banach space and $T = (T(t))_{t \geq 0}$ a C_0 -semigroup on X . Its *sun dual* is

$$X^\odot = \left\{ x' \in X' : \lim_{t \downarrow 0} \|T(t)'x' - x'\| = 0 \right\}.$$

It is a norm-closed $T(t)'$ -invariant subspace of X' .

If (x_n) is a Schauder basis, write x'_n for its biorthogonal functionals and

$$P_N x = \sum_{n=1}^N x'_n(x) x_n, \quad K = \sup_N \|P_N\| < \infty.$$

Proof Intuition

There are two independent mechanisms. First, any Schauder basis supports a bounded diagonal heat-type semigroup: multiply its n -th basis vector by e^{-nt} . Monotonicity of these scalars lets Abel summation control the operator norm even when the basis is conditional. The eigenvalues $-n$ force the generator to be unbounded.

Second, every coordinate functional is an eigenvector of the adjoint semigroup and hence belongs to the sun dual. Thus the sun dual is an infinite-dimensional subspace of $Y' \simeq \ell^1$. A gliding-hump argument shows that every such subspace contains a complemented copy of ℓ^1 . Dualizing that complemented summand puts a complemented copy of ℓ^∞ , and therefore c_0 , inside $(Y^\odot)'$.

The proof never requires the Bourgain–Delbaen basis to be unconditional or shrinking.

A Complemented ℓ^1 Lemma

We include the classical structural fact needed in the proof so that the only external dependency is the identification of the Bourgain–Delbaen space.

Lemma 1 (Complemented ℓ^1 -saturation). *Every infinite-dimensional closed subspace E of ℓ^1 contains a closed subspace $W \simeq \ell^1$ that is complemented in E . In fact, W may be chosen complemented in the ambient ℓ^1 .*

Proof. Let $R_N : \ell^1 \rightarrow \ell^1$ be truncation to the first N coordinates. Choose positive numbers (ε_k) with $\delta := 2 \sum_k \varepsilon_k < 1$. Inductively choose integers $0 = N_0 < N_1 < N_2 < \dots$ and unit vectors $z_k \in E$ such that

$$R_{N_{k-1}} z_k = 0, \quad \|(\text{Id} - R_{N_k}) z_k\|_1 < \varepsilon_k.$$

This is possible because $R_{N_{k-1}}|_E$ has finite rank, so its kernel in E is infinite-dimensional, and every vector of ℓ^1 has small tails.

Put

$$u_k = (R_{N_k} - R_{N_{k-1}}) z_k, \quad v_k = \frac{u_k}{\|u_k\|_1}.$$

The vectors v_k have pairwise disjoint supports and norm one, and

$$\|z_k - v_k\|_1 \leq \|z_k - u_k\|_1 + |1 - \|u_k\|_1| < 2\varepsilon_k.$$

Define $U : \ell^1 \rightarrow \ell^1$ by $U(a) = \sum_k a_k v_k$. Disjointness gives $\|U(a)\|_1 = \|a\|_1$. For every k , choose a norm-one functional ϕ_k supported on the support of v_k and satisfying $\phi_k(v_k) = 1$. Then

$$Px = \sum_k \phi_k(x) v_k$$

defines a contractive projection of ℓ^1 onto $V := U(\ell^1)$.

Define $S : \ell^1 \rightarrow \ell^1$ by $S(a) = \sum_k a_k z_k$. The preceding estimate gives

$$\|S - U\| \leq \sum_k \|z_k - v_k\|_1 < \delta < 1.$$

Consequently S is an isomorphic embedding. Moreover

$$B := U^{-1}PS : \ell^1 \rightarrow \ell^1$$

satisfies $\|B - \text{Id}\| < 1$, and hence is invertible. The operator

$$Q := SB^{-1}U^{-1}P : \ell^1 \longrightarrow S(\ell^1)$$

is bounded and satisfies $Q(Sa) = Sa$ for every $a \in \ell^1$. It is therefore a projection onto $W := S(\ell^1) = \overline{\text{span}}\{z_k : k \in \mathbb{N}\}$. Since every z_k lies in E , we have $W \subset E$; restricting Q to E proves complementability in E as well. $\square$

The General Diagonal-Semigroup Theorem

Theorem 1. *Let X be an infinite-dimensional Banach space with a Schauder basis and suppose $X' \simeq \ell^1$. Then X admits a C_0 -semigroup T with unbounded generator such that*

$$(X^\odot)' \text{ contains a complemented copy of } \ell^\infty.$$

In particular, $(X^\odot)'$ contains c_0 .

Proof. Let (x_n) be a Schauder basis, with biorthogonal functionals (x'_n) and basis projections P_N . For $t \geq 0$, define on basis vectors

$$T(t)x_n = e^{-nt}x_n.$$

For $t > 0$, Abel summation gives the operator-norm convergent formula

$$T(t) = \sum_{n=1}^{\infty} (e^{-nt} - e^{-(n+1)t}) P_n.$$

Indeed, the N -th finite Abel sum has the additional remainder $e^{-Nt}P_N$, whose norm tends to zero. Since all displayed coefficients are nonnegative,

$$\|T(t)\| \leq K \sum_{n=1}^{\infty} (e^{-nt} - e^{-(n+1)t}) = Ke^{-t}.$$

Set $T(0) = \text{Id}$. The semigroup law holds on every x_n , hence on the dense finite-support span and then on all of X . On that finite-support span, $T(t)x \rightarrow x$ as $t \downarrow 0$. The uniform bound $\sup_{0 < t \leq 1} \|T(t)\| \leq K$, followed by approximation with $P_N x$, proves strong continuity at zero. Thus T is a C_0 -semigroup.

Let A be its generator. For every n ,

$$\lim_{t \downarrow 0} \frac{T(t)x_n - x_n}{t} = -nx_n,$$

so $x_n \in D(A)$ and $Ax_n = -nx_n$. Hence A is unbounded.

For each coordinate functional,

$$T(t)'x'_n = e^{-nt}x'_n,$$

and therefore $x'_n \in X^\odot$. Since the x'_n are linearly independent, $X^\odot$ is infinite-dimensional. It is a closed subspace of $X' \simeq \ell^1$, so Lemma 1 supplies a complemented subspace $W \subset X^\odot$ with $W \simeq \ell^1$. Write $X^\odot = W \oplus Z$. Dualizing the direct sum yields

$$(X^\odot)' \simeq W' \oplus Z', \quad W' \simeq \ell^\infty.$$

Thus $(X^\odot)'$ contains a complemented copy of ℓ^∞ , which in turn contains its canonical copy of c_0 . $\square$

Application to the Bourgain–Delbaen Question

Tarbard’s detailed exposition of the original Bourgain–Delbaen construction shows that its biorthogonal vectors form a normalized Schauder basis [2, Chapter 2, especially the discussion following Proposition 2.4.3]. The same source records that every original Bourgain–Delbaen space in the class $\mathcal{Y}$, including the space denoted Y in the question, has dual isomorphic to ℓ^1 [2, Corollary 2.5.7].

Corollary 1. *The Bourgain–Delbaen space Y in Remark 4.15(2) of arXiv:2605.12121 admits a C_0 -semigroup with unbounded generator for which*

$$(Y^\odot)' \supset \ell^\infty \supset c_0,$$

where the copy of ℓ^∞ is complemented. Hence the question in that remark has an affirmative answer.

Proof. Apply Theorem 1 to the normalized Bourgain–Delbaen basis. Explicitly, if that basis is (y_n) , use

$$T(t)y_n = e^{-nt}y_n.$$

$\square$

Verification Notes

1. **Conditionality causes no problem.** The multiplier bound uses only $K = \sup_N \|P_N\| < \infty$ and the positive variation of $(e^{-nt})_n$. No unconditionality hypothesis is present.
2. **Shrinking is not used.** The proof needs only the individual coordinate functionals x'_n in the sun dual, not that they span all of X' .
3. **The generator is genuinely unbounded.** The eigenvalue equation $Ax_n = -nx_n$ rules out a bounded generator directly.

4. **Isomorphic, not isometric, duality is enough.** An isomorphism $X' \rightarrow \ell^1$ sends $X^\odot$ to an infinite-dimensional closed subspace; complementability and the isomorphism type of the extracted ℓ^1 copy pull back.
5. **The conclusion is stronger than requested.** The dual of the sun dual contains complemented ℓ^∞ , rather than merely a copy of c_0 .

Novelty Search and Limitations

On July 19, 2026, the lightweight indexes for run `fa.banach_001` were searched for arXiv:2605.12121, its title, maximal-regularity terms, input-to-state-stability terms, and the combinations “Bourgain–Delbaen”, “sun dual”, and “Lotz property”. No matching prior solution or attempt was present. Bounded web/arXiv searches used the exact source title and combinations of “Bourgain–Delbaen space”, “ C_0 -semigroup”, “sun dual”, “unbounded generator”, and “Lotz property”. They found the source and background construction papers but no later explicit answer. The arXiv record was still version 1, submitted May 12, 2026.

This search is not a global novelty certificate, and indexing can lag. The main review risk is bibliographic: an older general observation about diagonal semigroups on ℓ^1 -preduals may imply the same result without having been phrased as an answer to this recent question. The mathematical proof above is independent of that novelty issue.

The result answers the Bourgain–Delbaen residual question only. It does not settle the separate general equivalence problem for L^∞ -ISS and L^∞ -iISS in Remark 4.7 of the source.

Human-Review Recommendation

The candidate should be sent to a Banach-space or operator-semigroup expert. The most useful checks are: (i) matching the source’s Y with Tarbard’s class $\mathcal{Y}$; (ii) the gliding-hump projection in Lemma 1; and (iii) confirmation that “non-trivial” is intended in the unbounded-generator sense suggested by the Lotz-property discussion.

References

- [1] P. Preussler and F. L. Schwenninger, *Implications of structured continuous maximal regularity*, arXiv:2605.12121v1, 2026.
- [2] M. Tarbard, *Operators on Banach Spaces of Bourgain–Delbaen Type*, arXiv:1309.7469, 2013.
- [3] J. Bourgain and F. Delbaen, *A class of special $\mathcal{L}_\infty$ spaces*, Acta Math. 145 (1980), 155–176.